

POGRANICHNOYE, Russian Federation

WMO: 320760

Lat: 50.400N

Lon: 143.767E

Elev: 26

StdP: 14.68

Time Zone: 11.00 (E11)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.3	-14.4	-23.5	1.5	-16.8	-20.1	1.8	-12.6	24.3	21.8	20.9	22.6	4.7	280	0.737

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.7	72.1	61.5	67.2	59.5	63.3	57.7	63.4	69.5	60.8	65.6	58.6	62.5	7.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.0	80.1	65.5	58.5	73.4	63.0	56.5	68.1	60.5	28.9	70.1	27.1	65.8	25.5	62.5	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.0	14.4	12.0	-25.3	84.3	4.0	5.1	-28.2	88.0	-30.6	91.0	-32.9	93.8	-35.8	97.5		
(5)			-25.4	69.8	4.1	2.9	-28.4	71.9	-30.8	73.6	-33.1	75.2	-36.1	77.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	30.7	5.7	5.2	15.4	28.7	37.2	45.3	51.0	54.5	51.0	39.5	23.3	9.9		
	DBStd	19.16	10.18	9.57	8.71	6.36	6.06	6.36	5.23	5.41	5.65	5.83	9.49	9.87		
	HDD50	7400	1375	1253	1073	639	403	171	47	12	53	328	802	1244		
	HDD65	12534	1840	1673	1538	1089	862	593	437	332	419	790	1252	1709		
	CDD50	350	0	0	0	0	5	31	77	150	84	3	0	0		
	CDD65	9	0	0	0	0	0	2	2	5	0	0	0	0		
	CDH74	108	0	0	0	0	2	29	19	48	10	0	0	0		
	CDH80	23	0	0	0	0	0	9	5	8	1	0	0	0		
	WSAvg	5.1	6.1	5.5	5.5	5.1	4.9	4.5	3.9	3.9	4.5	5.5	5.7	6.1		
	PrecAvg	33.3	2.0	1.3	2.1	3.0	2.8	2.4	2.2	3.7	3.9	3.9	3.4	2.5		
(15) Precipitation	PrecMax	47.2	4.5	3.5	5.0	9.3	7.0	5.3	5.2	10.2	12.4	11.3	10.0	8.3		
	PrecMin	20.2	0.3	0.3	0.2	0.2	0.4	0.4	0.1	0.9	1.1	0.4	0.6	0.1		
	PrecStd	6.9	1.3	0.8	1.4	2.0	1.7	1.3	1.6	2.0	2.6	2.5	2.3	1.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.8	30.1	38.9	54.8	67.2	77.7	75.7	80.2	74.7	64.3	46.8	34.0		
		MCWB	29.9	28.1	32.4	43.2	52.3	60.3	64.4	66.1	60.1	52.1	40.5	32.9		
	2%	DB	29.2	27.7	33.1	46.0	58.3	65.4	66.6	72.4	69.2	57.8	41.7	31.5		
		MCWB	28.4	26.0	29.7	38.7	48.0	56.2	61.4	63.4	59.0	49.2	38.2	30.9		
	5%	DB	27.4	25.1	30.8	40.3	51.2	58.8	61.8	67.2	64.3	52.8	39.1	29.1		
		MCWB	26.4	23.8	28.6	35.2	44.9	53.5	58.4	61.1	57.0	46.7	35.7	28.2		
	10%	DB	23.4	21.5	29.0	36.6	46.3	54.5	58.6	63.2	60.5	49.5	36.3	26.3		
		MCWB	22.2	20.0	27.2	33.4	42.1	50.8	56.0	59.4	55.2	45.5	33.7	24.9		
	0.4%	WB	30.6	28.7	33.4	44.5	53.7	61.8	65.6	68.2	63.2	53.6	42.2	33.2		
		MCDB	30.7	29.9	36.7	52.6	64.0	74.7	72.9	75.7	69.8	61.3	44.3	33.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	28.6	26.4	30.8	39.0	48.8	57.2	61.7	64.4	60.4	50.7	39.5	30.9		
		MCDB	29.0	27.5	32.3	45.3	56.3	64.1	66.2	70.9	66.3	55.2	41.4	31.3		
	5%	WB	26.3	23.8	29.1	35.8	45.4	53.9	58.8	61.9	58.0	48.3	36.4	28.4		
		MCDB	27.4	25.1	30.4	39.5	50.8	58.2	61.8	66.5	63.2	51.5	38.4	29.0		
	10%	WB	22.3	20.2	27.3	33.7	42.2	51.0	56.3	59.6	55.9	46.1	34.0	25.0		
		MCDB	23.4	21.4	28.8	36.0	46.0	54.1	58.4	63.0	59.8	48.8	36.0	26.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.2	19.7	18.3	12.3	10.8	11.2	10.0	11.7	15.4	16.1	14.9	16.0		
		MCDBR	11.8	15.5	18.5	21.5	23.3	22.4	18.3	21.1	23.0	23.1	16.3	11.6		
		MCWBR	11.7	14.5	15.4	14.1	14.3	14.2	13.0	13.4	13.2	15.5	12.9	10.9		
		MCDBR	11.4	14.0	15.6	19.3	22.1	20.4	17.3	19.3	19.4	20.2	13.7	9.8		
	5% WB	MCWBR	11.6	13.3	13.6	13.1	14.5	13.6	12.8	13.2	12.4	14.8	11.7	10.0		
(40) Clear-Sky Solar Irradiance	taub		0.285	0.313	0.379	0.471	0.502	0.466	0.448	0.422	0.377	0.385	0.348	0.298		
	taud		2.061	1.984	1.825	1.693	1.851	2.134	2.253	2.343	2.407	2.190	2.123	2.088		
	Ebn at Noon		226	251	252	241	241	253	256	258	258	224	198	200		
	Edh at Noon		27	37	54	69	62	47	41	36	30	31	25	23		
(44) All-Sky Solar Radiation	RadAvg		415	771	1171	1413	1536	1672	1512	1316	1088	655	353	282		
	RadStd		41	60	86	142	156	166	186	113	90	56	40	34		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	+0.67	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page